

3rdParty MCAL Integration

Technical Reference

Cypress Cyt2xy

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Authors	virova
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Reference Documents

No.	Source	Title	Version
[1]	Vector	UserManual_3rdParty-MCAL-Integration.pdf	see delivery
[3]	Vector	ScreenCast_McalIntegration_Tresos.pdf	see delivery

Scope of the Document

This document contains information about the integration of 3rd Party MCAL into Vector software stack.

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1 MCAL Integration

1.1 Type of Integration

Basic Integration

Both configuration tools, EB tresos™ as well as Vector DaVinci Configurator, are used for configuration.

Recommended workflow:

Start initial configuration with EB tresos™, export it in AUTOSAR format and import it into Vector DaVinci Configurator. Generation and minor changes in configuration are done in Vector DaVinci Configurator.

For usage with Vector DaVinci Configurator 5 please refer to chapter 'Mixed configuration tool usage' in the document `UserManual_3rdParty-MCAL-Integration.pdf` [1].



Multimedia Link

For additional information please also refer to the Vector screen cast referenced in `ScreenCast_McalIntegration_Tresos.pdf` [3]

1.2 MCAL Location within SIP

The 3rd Party MCAL is separated from the Vector parts within the SIP. It might not even be part of the delivery. Please refer to chapter 'First Steps' in document `UserManual_3rdParty-MCAL-Integration.pdf` [1] after performing the following steps.

1.3 Preparation of 3rd Party Packages

These steps are necessary due to the specific installation process of the Cypress Software packages:

- > Run the installation routines of the EB tresos™ and the 3rd party MCAL package (you can leave the default paths).
- > Install also the stub plugins for Dem, EcuM and Os even if you do not intend to use the MCAL stubs in EB tresos™. They are needed to solve references in the generation process.
- ▶ Now the 3rd party packages are prepared for usage with the Vector delivery.

**Note**

In UserManual_3rdParty-MCAL-Integration.pdf [1] you will find a detailed description of the recommended workflow including usage of EB tresos™.

**Caution**

For the case that your Vector SIP contains communication components like Can, Lin, Fr, please do not use these components from the MCAL.

1.4 Supported 3rd Party Products

This integration supports the following Cypress targets:

1.4.1 TraveoII

- ▶ ProdRel_WW2210_SW-MCAL-TraveoII_v1.14.0
- ▶ ProdRel_WW2144_SW-MCAL-TraveoII_V1.13.0
- ▶ ProdRel_WW2139_SW-MCAL-TraveoII_v1.12.0
- ▶ ProdRel_WW2130_SW-MCAL-TraveoII_v1.11.0
- ▶ ProdRel_WW2122_SW-MCAL-TraveoII_v1.10.0
- ▶ ProdRel_WW2113_SW-MCAL-TraveoII_v1.9.0
- ▶ ProdRel_WW2051_SW-MCAL-TraveoII_v1.8.2
- ▶ BetaRel_WW2038_SW-MCAL-TraveoII_v1e.8.0
- ▶ ProdRel_WW2033_SW-FEE-TraveoII_v1.7.0
- ▶ ProdRel_WW2029_SW-MCAL-TraveoII_v1.6.0
- ▶ ProdRel_WW2018_SW-MCAL-TraveoII_v1.5.0
- ▶ ProdRel_WW2004_SW-MCAL-TraveoII_v1.4.0
- ▶ BetaRel_WW1934_SW-MCAL-TraveoII_v1e.3.1
- ▶ BetaRel_WW1934_SW-MCAL-TraveoII_v1e.3.0
- ▶ ProdRel_WW1929_SW-MCAL-TraveoII_v1.2.4
- ▶ BetaRel_WW1912_MCAL_IARSupport_TraveoII_v1e.0.0
- ▶ BetaRel_WW1910_MCAL_TraveoII_v1e.2.2
- ▶ BetaRel_WW1905_MCAL_TraveoII_v1e.2.1

**Note**

Please refer to the Release Notes of the 3rd Party Products for further information, e.g. regarding supported versions, derivatives and compilers.

**Note**

Only official 3rd Party vendor releases are part of this Vector integration package. Therefore, any customer-specific releases cannot be considered.

**Caution**

To find out if there are further Hotfixes available for your MCAL package, please contact the 3rdParty Vendor.

It is essential to replace the affected module plugins in your original package before you start Script_MCAL_Prepare.bat.

1.5 Configuration Tools

- ▶ Vector DaVinci Configurator 5
- ▶ EB tresos™

2 Vector Comment

The user should consider the attached `UserManual_3rdParty-MCAL-Integration.pdf` [\[1\]](#) for further information regarding Vector integration and setup of a project.

2.1 Known Issues

The MCAL package in use might not be the latest one. Updates or Hot Fixes might be available from the vendor.

If you discover errors in the MCAL during development, or suspect that faulty behavior could be caused by the MCAL, please contact the 3rdParty Vendor. If there are updates for your MCAL available, it might be helpful to use them. The corresponding 3rdParty Vendor Release Notes will tell you which errors have been fixed.

The errors documented in the following chapters were detected at Vector during the MCAL integration and reported to the 3rdParty Vendor.



Note

Vector makes every effort to ensure that this integration is compatible with all MCAL packages of the corresponding Controller Family as long as the 3rdParty Vendor does not change the structure of the packages. However, not all MCAL packages can be tested and therefore the user may find errors either in Vector's files or in the MCAL that are not listed in this document.



Caution

Modifications to the MCAL, which may be described in the following chapters under the keyword 'Workaround', must take place after the 3rdPartyMCALIntegrationHelperTool has run. Otherwise the changes will be overwritten.



Note

Necessary patches are provided via the SIP folder
ThirdParty\<Short_Name>\VectorIntegration\Patches\ (called 'patch folder' in this chapter).

2.1.1 Traveoll

2.1.1.1 BetaRel_WW1905_MCAL_Traveoll_v1e_2_2

2.1.1.1.1 File TSAtomic_Lib_Arch_Asm_GHS.s not found.

Below error occurs while compiling the MCAL.

make: *** No rule to make target

'obj/external/ThirdParty/Mcal_Cyt2xy/Supply/tresos/plugins/Base_TS_T40D13M1I0R0/lib_src/CYT2/TSAtomic_Lib_Arch_Asm_GHS.o', needed by 'TestSuit.elf'. Stop.



Workaround

In external\BSW\Mcal_Cyt2xy\mak\Mcal_Cyt2xy_rules.mak

Comment the below line

```
ASM_FILES_TO_BUILD +=
```

```
$(BASE_PATH)\$(BASE_PLUGIN)\lib_src\CYT2\TSAtomic_Lib_Arch_Asm_GH  
S.s
```

2.1.1.2 BetaRel_WW1910_MCAL_Traveoll_v1e.2.2

2.1.1.2.1 File TSAtomic_Lib_Arch_Asm_GHS.s not found.

See chapter 2.1.1.1.1.

2.1.1.3 BetaRel_WW1912_MCAL_IARSupport_Traveoll_v1e.0.0

2.1.1.3.1 File TSAtomic_Lib_Arch_Asm_GHS.s not found.

See chapter 2.1.1.1.1.

2.1.1.4 ProdRel_WW2051_SW-MCAL-Traveoll_v1.8.2

2.1.1.4.1 Generation error of Mcu in Configurator 5

After a restart of the DaVinci Configurator 5, the generation of Mcu leads to the following error message:

```
ERROR 20-12-21,07:13:57 (1803) Invalid XPath-expression for Attribute "ENABLE" of node  
"McuScgPllSettings": (35016) Cannot get a resource file for target MXS40 and derivate  
CYT3B0. Maybe this is due to module restrictions.
```

This issue has been reported to Cypress with the case number: #00606320



Workaround

After each restart of DaVinci Configurator 5, remove the Mcu module and import the configuration again.

2.1.1.5 ProdRel_WW2113_SW-MCAL-Traveoll_v1.9.0

2.1.1.5.1 Generation error of Mcu in Configurator 5

See chapter 2.1.1.4.1

2.1.1.6 ProdRel_WW2122_SW-MCAL-TraveoII_v1.10.0**2.1.1.6.1 Generation error of Mcu in Configurator 5**

See chapter 2.1.1.4.1

2.1.1.7 ProdRel_WW2130_SW-MCAL-TraveoII_v1.11.0**2.1.1.7.1 Generation error of Mcu in Configurator 5**

See chapter 2.1.1.4.1

2.1.1.8 ProdRel_WW2139_SW-MCAL-TraveoII_v1.12.0**2.1.1.8.1 Generation error of Mcu in Configurator 5**

See chapter 2.1.1.4.1

2.1.1.9 ProdRel_WW2144_SW-MCAL-TraveoII_V1.13.0**2.1.1.9.1 Generation error of Mcu in Configurator 5**

See chapter 2.1.1.4.1

2.1.1.10 ProdRel_WW2210_SW-MCAL-TraveoII_v1.14.0**2.1.1.10.1 Generation error of Mcu in Configurator 5**

See chapter 2.1.1.4.1

2.2 Restrictions

2.2.1 General

2.2.1.1 Missing includes Fls_Cfg.h and Fls_PBCfg.h in Fls.h

Below error occurs while compiling EcuM_Callout_Stubs.c

```
CC      EcuM_Callout_Stubs.o
```

```
"../../../../internal/StartApplication/Appl/Source/EcuM_Callout_Stubs.c", line 269: error #20:
    identifier "Fls_Config_0" is undefined
    Fls_Init( &Fls_Config_0 );
```

This issue has been reported to Cypress with the case number: **00473401**



Workaround

Follow the instructions in Cypress' UsersGuide_Fls_1_0_TraveoII_Autosar.pdf, chapter 2.3 "Adapting Your Application"

2.2.1.2 Fls: FlsSystemcallCalloutFunction mandatory in DaVinci

While the parameter `FlsSystemcallCalloutFunction` can be disabled in EB tresos, it is mandatory in DaVinci Configurator 5.

If the parameter is disabled in EB tresos, the generation result in `Fls_Cfg.h` is:

```
#define FLS_SYSTEMCALL_CALLOUT_FUNCTION NULL_PTR
```

If you like to get the same result with DaVinci Configurator 5, it must be configured to `NULL_PTR` there.

2.2.1.3 All the macros for the different MCU_ResetReasons are given the same value in Mcu_Cfg.h

While generating Mcu module in DaVinci Configuration, In the file `Mcu_Cfg.h` all the different macros for `McuResetReason` are given a single value `0U` as below.

```
/* Generate symbolic names for the different reset reason settings */
#define McuConf_McuResetReasonConf_MCU_ACT_FAULT_RESET 0U
#define McuConf_McuResetReasonConf_MCU_BOD_VCCD_RESET 0U
#define McuConf_McuResetReasonConf_MCU_BOD_VDDA_RESET 0U
#define McuConf_McuResetReasonConf_MCU_BOD_VDDD_RESET 0U
#define McuConf_McuResetReasonConf_MCU_CSV_HF_RESET 0U
#define McuConf_McuResetReasonConf_MCU_CSV_REF_RESET 0U
```

```
#define McuConf_McuResetReasonConf_MCU_DPSLP_FAULT_RESET 0U
#define McuConf_McuResetReasonConf_MCU_MCWDT0_RESET 0U
#define McuConf_McuResetReasonConf_MCU_MCWDT1_RESET 0U
#define McuConf_McuResetReasonConf_MCU_MCWDT2_RESET 0U
#define McuConf_McuResetReasonConf_MCU_MCWDT3_RESET 0U
#define McuConf_McuResetReasonConf_MCU_OCD_ACTIVE_REGULATOR_RESET
0U
#define
McuConf_McuResetReasonConf_MCU_OCD_DEEPSLEEP_REGULATOR_RESET 0U
#define McuConf_McuResetReasonConf_MCU_OVD_VCCD_RESET 0U
#define McuConf_McuResetReasonConf_MCU_OVD_VDDA_RESET 0U
#define McuConf_McuResetReasonConf_MCU_OVD_VDDD_RESET 0U
#define McuConf_McuResetReasonConf_MCU_POWER_ON_RESET 0U
#define McuConf_McuResetReasonConf_MCU_PXRES_RESET 0U
#define McuConf_McuResetReasonConf_MCU_REGHC_OCD_RESET 0U
#define McuConf_McuResetReasonConf_MCU_REGHC_PMIC_RESET 0U
#define McuConf_McuResetReasonConf_MCU_RESET_UNDEFINED 0U
#define McuConf_McuResetReasonConf_MCU_STRUCTURAL_XRES_RESET 0U
#define McuConf_McuResetReasonConf_MCU_SW_RESET 0U
#define McuConf_McuResetReasonConf_MCU_TEST_DEBUG_RESET 0U
#define McuConf_McuResetReasonConf_MCU_WAKEUP_RESET 0U
#define McuConf_McuResetReasonConf_MCU_WATCHDOG_RESET 0U
#define McuConf_McuResetReasonConf_MCU_XRES_RESET 0U
```



Workaround

Enable both “Using MCAL Pre Configuration” and “Performance Optimization” for the generation to use the MCAL pre-configuration for the reset reasons.

2.2.1.4 Fatal error: 'ErrorCalloutHandler.h' file not found

The below error occurs while compiling the MCAL.

In file included from

../../external/ThirdParty/Mcal_Cyt2xy/Supply/tresos/plugins/Dio_TS_T40D13M1I0R0/src/
Dio_ErrorHandler.c:27:

GenData/include/Dio_Cfg_Include.h:27:10: fatal error: 'ErrorCalloutHandler.h' file not found

#include <ErrorCalloutHandler.h>

1 error generated.

make: *** [..\..\MakeSupport\Global.Makefile.target.4.mk:1000:

obj/external/ThirdParty/Mcal_Cyt2xy/Supply/tresos/plugins/Dio_TS_T40D13M1I0R0/src/Dio_ErrorHandler.o] Error 1

make: *** Waiting for unfinished jobs....

Depend Adc.d

In file included from

../../external/ThirdParty/Mcal_Cyt2xy/Supply/tresos/plugins/Adc_TS_T40D13M1I0R0/src/Adc.c:30:

GenData/include/Adc_ExternalInclude.h:29:10: fatal error: 'ErrorCalloutHandler.h' file not found

#include <ErrorCalloutHandler.h>



Workaround

Copy the file ErrorCalloutHandler.h from path VectorIntegration\Patches to the include folder \StartApplication\App\Include

2.2.1.5 Usage of the different compiler vendors

Wdg generates different assembler files for IAR and GHS compiler. Depending on the switch "COMPILER_MANUFACTURER", the right one is chosen in Mcal_Cyt2xy_rules.mak:

```
250 ifneq ($(MCAL_EXCLUDE_WDG),1)
251 GENERATED_SOURCE_FILES += $(GENDATA_DIR)\src\Wdg*.c
252 ifeq ($(COMPILER_MANUFACTURER),GHS)
253 ASM_GENERATED_SOURCE_FILES += $(GENDATA_DIR)\src\Wdg_66_IA_Trigger_Asm_GHS.s
254 endif
255 ifeq ($(COMPILER_MANUFACTURER),IAR)
256 ASM_GENERATED_SOURCE_FILES += $(GENDATA_DIR)\src\Wdg_66_IA_Trigger_Asm_IAR.s
257 endif
258 endif
```

The switch is usually set in "Makfile.static" if Vector MakeSupport is used. If it is not set in the make file, default compiler manufacturer is GHS:

```
79
80 #default compiler version if none is set in make file
81 ifeq ($(COMPILER_MANUFACTURER),)
82 COMPILER_MANUFACTURER = GHS
83 endif
```

2.2.1.5.1 BetaRel_WW1912_MCAL_IARSupport_TraveoII_v1e.0.0

For MCAL version BetaRel_WW1912_MCAL_IARSupport_TraveoII_v1e.0.0, additionally the folders

- Can_TS_T40D13M1I0R0
- Make_TS_T40D13M1I0R0
- Wdg_TS_T40D13M1I0R0

in the path ..\tresos\plugins of the MCAL BetaRel_WW1910_MCAL_TraveoII_v1e_2_2 must be replaced by the equivalent folders of the MCAL BetaRel_WW1912_MCAL_IARSupport_TraveoII_v1e.0.0.

3 Glossary and Abbreviations

3.1 Glossary

Term	Description
3 rd party components / MCAL	BSW modules not provided by Vector. Vector may have integrated the software within the SIP but does not take over any responsibility regarding functionality of these modules.
DaVinci Configurator	Configuration and generation tool for Vector MICROSAR components

Table 3-1 Glossary

3.2 Abbreviations

Abbreviation	Description
MCAL	Microcontroller Abstraction Layer
AUTOSAR	Automotive Open System Architecture
SIP	Software Integration Package (as provided by Vector)
Msn	Module Short Name according AUTOSAR

Table 3-2 Abbreviations

4 Contact

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www.vector.com